

UCI Online Retail — Customer Segmentation

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1 Project objective

Build a reproducible unsupervised customer-segmentation pipeline on the UCI Online Retail dataset (id 352): one year of transactions from a UK-based online gift retailer. The statistical goal is to map raw transaction lines into customer-level behavioural features, reduce feature redundancy via PCA, cluster customers in the reduced space, validate the clustering, and persist a deployable scoring artefact. The methodological scope is restricted to ISLR Ch. 12 (PCA, k-means, hierarchical clustering); ensemble methods, kernel methods, or generative models are out of scope.

2 Data source and cleaning

The raw download contains 541,909 transaction lines covering 8 columns: `InvoiceNo`, `StockCode`, `Description`, `Quantity`, `InvoiceDate`, `UnitPrice`, `CustomerID`, `Country`. The cleaning pipeline (`src/python/data_prep.py`):

1. drops rows missing `CustomerID`;
2. removes duplicate rows;
3. removes non-product `StockCodes` (`POST`, `D`, `C2`, `M`, `BANK CHARGES`, `AMAZONFEE`, `DOT`, `CRUK`, `PADS`, `B`);
4. **separates** cancellation invoices (`InvoiceNo` starting with `C`) into a sibling table rather than discarding them;
5. retains positive-quantity, positive-price lines in the main table;
6. standardises the customer key as `CustomerID` everywhere.

After cleaning: ~391,150 positive transactions across ~4,334 customers, plus a ~9,000-row cancellations table that drives the net-of-returns features below.

3 Feature engineering

For each customer i we construct

$$\mathbf{x}_i = (R_i, F_i, M_i^{\text{gross}}, M_i^{\text{net}}, Q_i, U_i, A_i, P_i, T_i, C_i, V_i, r_i, \rho_i, S_i)$$

with the definitions in the table below. Let $T_{\max}$ be the maximum positive-transaction date.

Symbol	Feature	Definition
R_i	Recency	$T_{\max} + 1 - \max\{t : t \in \text{tx}_i\}$ (days)
F_i	Frequency	number of distinct positive invoices for customer i
M_i^{gross}	GrossMonetary	$\sum_{j \in i} q_j p_j$ over positive lines
V_i	ReturnValueAbs	$ \sum_{j \in \text{cancel}_i} q_j p_j $
M_i^{net}	NetMonetary	$M_i^{\text{gross}} - V_i$
Q_i	TotalQuantity	$\sum_{j \in i} q_j$ over positive lines
U_i	UniqueProducts	$ \{\text{StockCode}_j : j \in i\} $
A_i	AvgOrderValue	mean invoice total
P_i	AvgUnitPrice	$\sum_j q_j p_j / \sum_j q_j$ (quantity-weighted)
T_i	TenureDays	$\text{LastPurchase}_i - \text{FirstPurchase}_i$
C_i	ReturnInvoices	count of cancellation invoices
r_i	ReturnRate	$C_i / (F_i + C_i)$
ρ_i	ReturnValueRate	V_i / M_i^{gross}
S_i	SinglePurchase	$\mathbf{1}\{F_i = 1\}$

AvgUnitPrice is **quantity-weighted** rather than the mean of line-level **UnitPrice**, since the line-level mean overweights tiny low-priced items. We additionally winsorise:

$$P_i^{\text{winsor}} = \min(P_i, Q_{0.99}(P)).$$

Right-skewed features get the standard log transform $\tilde{x} = \log(1 + x)$; for **NetMonetary**, which can be slightly negative when cancellations exceed gross sales, we use the sign-preserving $\tilde{x} = \text{sgn}(x) \log(1 + |x|)$.

4 Exploratory data analysis

Three EDA findings drove the modelling choices.

Heavy right-skew on monetary. The top ~1% of customers control roughly a quarter of net revenue; the top 5% control around half. This is documented as a “top-customer revenue concentration curve” rather than a Lorenz curve (a true Lorenz curve sorts ascending). The log transform reduces but does not eliminate the skew — the upper shoulder of $\log(1 + M^{\text{net}})$ is real wholesale-leaning customers, kept in and addressed via sensitivity analysis rather than removed.

Severe collinearity. $M^{\text{gross}} \approx M^{\text{net}} \approx Q \approx F$ form a tight block under both Pearson and Spearman correlation. `AvgOrderValue` and `AvgOrderItems` carry essentially the same information; only `AvgOrderValue` is retained. Recency is largely independent of the spending block.

Geographic skew. UK $\approx 83\%$ of revenue. Including `Country` as a clustering input would force a near-binary split that drowns out behavioural signal; `Country` is therefore excluded from clustering and re-introduced only as a profiling variable.

A separate finding: `StockCode 23843` (“PAPER CRAFT, LITTLE BIRDIE”) dominates product-level revenue, driven almost entirely by a single customer’s single bulk order. Per-cluster product profiles are quoted both with and without 23843; the report’s narrative uses the without-anomaly version.

5 PCA methodology and interpretation

The modelling matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ uses $p = 7$ features:

$$(\log(1 + R), \log(1 + F), \widetilde{\log} M^{\text{net}}, \log(1 + U), \log(1 + A), P^{\text{winsor}}, T).$$

We standardise column-wise to $\mathbf{Z}$ with zero mean and unit variance, compute the correlation matrix $\mathbf{S} = \frac{1}{n-1} \mathbf{Z}^\top \mathbf{Z}$, and find eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_p$ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_p$. The k -th principal component score for customer i is

$$\text{PC}_{ik} = \mathbf{z}_i^\top \mathbf{v}_k,$$

and the proportion of variance explained by component k is $\text{PVE}_k = \lambda_k / \sum_m \lambda_m$. We retain the smallest K such that $\sum_{k \leq K} \text{PVE}_k \geq 0.80$ (default $K = 3$).

PC signs are arbitrary in PCA: $\mathbf{v}_k$ and $-\mathbf{v}_k$ are equally valid, so interpretation is in terms of the *contrast* across features, not the absolute sign.

PC1 — engagement / customer value. Positive on $\log F$, $\widetilde{\log} M^{\text{net}}$, $\log U$, and T ; negative on $\log R$. High PC1 = recent, frequent, broad-product, longer-tenure customers. Low PC1 = lapsed one-time buyers.

PC2 — order-size vs. tenure contrast. Negative on $\log A$ and P^{winsor} ; positive on T and $\log F$. Low PC2 = wholesale-leaning behaviour (fewer, larger, higher-priced orders). PC2 is *not* purely a “tenure / lifecycle” axis; that description omits the `AvgOrderValue` contrast which is the more informative half.

PC3 — residual price-point. Once PC1 and PC2 absorb the engagement and order-size dimensions, PC3 picks up the remaining P^{winsor} signal — what tier of price the customer typically buys at.

6 Cluster selection

K-means on the PC scores minimises

$$\min_{C_1, \dots, C_K} \sum_{k=1}^K \sum_{i \in C_k} \|\mathbf{y}_i - \mu_k\|^2, \quad \mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{y}_i.$$

We sweep $k \in \{2, \dots, 10\}$ with `nstart` = 50 and a fixed seed, reporting three diagnostics:

- **Within-cluster sum of squares (elbow).** Strictly decreasing in k ; we look for a “kink” where the marginal gain flattens.
- **Mean silhouette width.** For each customer $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$ where $a(i)$ is the average distance to other points in customer i ’s cluster and $b(i)$ is the minimum average distance to points in another cluster. Higher $\bar{s}$ means clearer separation.
- **Gap statistic** (Tibshirani 2001). $\text{Gap}(k) = \mathbb{E}_n^*[\log W_k] - \log W_k$ compares observed within-cluster dispersion W_k to a uniform-noise reference. The **SE-max rule** picks the smallest k such that $\text{Gap}(k) \geq \text{Gap}(k+1) - \text{SE}_{k+1}$, biased toward parsimony.

The diagnostics agree broadly. Silhouette peaks at $k = 2$ and remains competitive at $k = 3$; the gap-statistic SE-max rule selects $k = 3$; the elbow is gradual. We adopt $k = 3$ as the final segmentation because (a) it is statistically defensible under the most rigorous diagnostic, (b) it produces interpretable high / mid / low value segments rather than the degenerate two-way split that $k = 2$ produces, and (c) the persona-level profiles (next section) are materially more actionable. $k = 2$ is retained as a sensitivity model.

7 Cluster validation

Bootstrap stability (Hennig 2007). Resample customers with replacement, refit, match each bootstrapped cluster back to its original via Jaccard similarity $J(A, B) = |A \cap B| / |A \cup B|$, and report the per-cluster mean across bootstraps. Jaccard ≥ 0.75 = stable; < 0.6 = noise. K-means at $k = 3$ produces all three Jaccards above ~ 0.87 — a strong result. Ward hierarchical clustering at the same k produces one stable cluster (~ 0.81) and two borderline ones ($\sim 0.59, 0.67$).

Adjusted Rand index (`mclust::adjustedRandIndex`) compares two partitions while adjusting for chance agreement; ARI = 1 means identical labels (up to relabelling), ARI = 0 means agreement no better than chance. We report two ARIs:

- k-means vs. Ward: quantifies the partial disagreement that the contingency table makes visible. The cross-tab is *not* cleanly block-diagonal — k-means prefers spherical equal-size clusters; Ward does not.
- k-means on raw vs. scaled PC scores: tests whether PC1’s larger variance (which weights it more heavily under Euclidean distance) is doing the segmentation alone. ARI close to 1 means the segmentation is robust to that preprocessing choice.

Because the project goal is *stable* customer segmentation, k-means is promoted as the final label source despite Ward’s marginally higher silhouette.

8 Final customer profiles

Cluster labels are re-assigned in descending order of median net spend so that A is always the highest-value group, C the lowest, regardless of the random seed.

Cluster	Approx. share of customers	Approx. share of net revenue	Description
A — Loyal high-value	$\sim 28\%$	$\sim 75\%$	Recent, frequent, broad product variety, long tenure. Possible wholesale-leaning behaviour.

Cluster	Approx. share of customers	Approx. share of net revenue	Description
B — Developing mid-value	~38%	~21%	Moderate spend and frequency, mid-tenure. Retention / upsell target.
C — Lapsed one-time	~34%	~4%	Single-purchase predominant ($S_i = 1$), long recency, low spend. Win-back or accept as low margin.

`TenureDays = 0` in cluster C does **not** mean “new customer” — the `SinglePurchase` flag confirms these are predominantly customers who bought exactly once and have not returned. They should be described as *lapsed*, not new.

A shallow `rpart` decision tree fit on the log-features recovers the cluster labels with high in-sample accuracy from a handful of axis-aligned thresholds on Recency, Frequency, and NetMonetary. This is a **descriptive surrogate model**, not a predictive one — the tree is fit and scored on the same data, so its accuracy is not evidence of out-of-sample generalisation. Its purpose is to show that the segmentation can be communicated to a non-statistical audience as a small set of RFM thresholds.

9 Software engineering and reproducibility

The repository is organised so that `make all` from a clean checkout reproduces every artefact:

```
make data    -> python src/python/data_prep.py
make model   -> Rscript scripts/run_pipeline.R
make reports -> Rscript scripts/render_reports.R
make test    -> testthat::test_dir("tests")
```

The pipeline is parameterised through `config.yml`: `paths`, the seven-feature modelling matrix, $K = 3$, `nstart = 50`, the bootstrap B , and the random seed all live there. Every R module sources `src/R/00_setup.R`, which loads the seed and packages once.

Data integrity is enforced at three layers: assertions inside `src/python/data_prep.py`, runtime checks in `src/R/01_validate_data.R`, and `testthat` tests in `tests/` that re-check the artefacts after the pipeline writes them.

The deployable artefact is `models/segmentation_pipeline.rds`, which carries the prcomp object (with center/scale vectors and rotation matrix), the k-means centroids, the cluster label map, the personas, the version, the timestamp, and the diagnostic numbers. The function `score_customers()` in `src/R/06_score_new_customers.R` uses it to project new customer feature vectors into PC space and assign each to its nearest centroid:

$$\mathbf{z}_{\text{new}} = \frac{\mathbf{x}_{\text{new}} - \boldsymbol{\mu}_{\text{train}}}{\sigma_{\text{train}}}, \quad \mathbf{y}_{\text{new}} = \mathbf{z}_{\text{new}} \mathbf{V}_{1:K}, \quad \text{cluster}(\mathbf{y}_{\text{new}}) = \arg \min_k \|\mathbf{y}_{\text{new}} - \boldsymbol{\mu}_k\|^2.$$

10 Limitations

- **Gross vs. net monetary.** The pipeline defaults to `NetMonetary`; earlier iterations used `GrossMonetary` and would assign customer 12346 (large gross + matching cancellation) to the

high-value segment. Switching `monetary_source` in `config.yml` and re-fitting is the recommended sensitivity check.

- **High-value tail.** A small number of customers in the top 1% of spend carry disproportionate weight in cluster centroids. The `SinglePurchase` flag and `ReturnValueAbs` profile diagnose whether their inclusion is appropriate.
- **StockCode 23843 anomaly.** A single bulk order skews any product-level interpretation that includes it. Product profiles in the report are quoted from the without-anomaly version.
- **One-year window.** Recency and tenure are bounded by the observation window; segments labelled “recent” may simply be customers who have not had time to lapse.
- **Descriptive, not causal.** Cluster membership is an unsupervised assignment. It does not predict future behaviour without further modelling, and the decision-tree approximation is in-sample only.

11 Recommended next steps

- Promote `NetMonetary` to the canonical revenue measure across all reports; retain `GrossMonetary` only as a sensitivity benchmark.
- Re-fit the pipeline excluding the top 1% of customers by spend; report whether the three-segment story survives.
- Compare against richer clusterers — Gaussian mixture models (`mclust::Mclust`) for soft assignments and BIC-based k selection, or PAM with Gower distance for handling `Country` natively.
- If a second period of transaction data becomes available, score the new transactions through `score_customers()` and check whether intra-segment behaviour remains coherent. That is the only true external validity check for an unsupervised segmentation.
- Translate segments into measurable marketing experiments — a retention test on cluster B with a defined upsell metric, or a win-back test on the most-recent slice of cluster C.

12 Conclusion

This project builds a reproducible unsupervised learning pipeline for customer segmentation. Raw transactions become customer-level RFM and behavioural features (with cancellations preserved separately so net monetary can be computed honestly), are projected into a lower-dimensional PCA space, and are clustered with k-means, validated by bootstrap stability and adjusted Rand against Ward hierarchical clustering as a robustness check. The final $k = 3$ segmentation separates customers into a high-value loyal core (cluster A), a developing mid-value middle (cluster B), and a low-value / lapsed one-time tail (cluster C). PCA identifies engagement / customer value as the dominant axis of variation; later components capture order-size and price-point contrasts.

The principal statistical finding is that customer value is highly concentrated — a relatively small group of recent, frequent, high-spending customers accounts for a large share of revenue. The segments are descriptive, interpretable, and operationally useful, but they depend on preprocessing choices documented in `models/model_card.md`. The pipeline is shipped as a reproducible repository — configuration files, validation checks, tests, saved model artefacts, and a scoring function for assigning future customers to the learned segments — not as a collection of disconnected notebooks.